MATH.APP.220 Multivariable Calculus

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The Riemann sum of a three variable function

- Let $T \subset \mathbb{R}^3$ be a bounded set. Cover T with a rectangular cuboid, whose sides are in the directions of the coordinate planes. Divide the cuboid into subcuboids T_{ijk}, $1 \leq i \leq m$, $1 \leq j \leq n$, $1 \leq k \leq p$, whose edges have lengths $\triangle x_i$, $\triangle y_j$ and $\triangle z_k$, and thus the volumes are $\triangle x_i \triangle y_j \triangle z_k$.
- Choose tag points $\left(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*\right) \in T_{ijk}$.
- The norm $|P|$ of the partition is the maximum over the diameters (=space diagonals) of all T_{ijk}.
- The **Riemann sum** of a bounded function f defined in the cuboid is

$$\sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{k=1}^{p} f\left(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*\right) \triangle x_i \triangle y_j \triangle z_k.$$

Definition of a triple integral

- Let $T \subset \mathbb{R}^3$ be a bounded set and $f : T \to \mathbb{R}$ a bounded function. Extend f by zero, i.e. define $f^0 : \mathbb{R}^2 \to \mathbb{R}$,

$$f^0(x,y,z) = \begin{cases} f(x,y,z), & \text{jos } (x,y,z) \in T, \\ 0, & \text{jos } (x,y,z) \notin T. \end{cases}$$

- The function f is **integrable** in T, if there exists the **integral**

$$\iiint_T f(x,y,z)\, dxdydz$$
$$= \lim_{|P| \to 0} \sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{k=1}^{p} f^0\left(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*\right) \triangle x_i \triangle y_j \triangle z_k.$$

Triple integral in a rectangular cuboid

- If the boundary of a closed and bounded set $T \subset \mathbb{R}^3$ consists of finitely many piecewise smooth surfaces and $f : T \to \mathbb{R}$ is a continuous function, then f is integrable in T.
- Fubini's theorem: The function $f : T \to \mathbb{R}$ that is continuous in the cuboid $T = [a_1, a_2] \times [b_1, b_2] \times [c_1, c_2]$, is integrable and

$$\begin{aligned}\iiint_T f(x,y,z)\,dxdydz &= \int_{a_1}^{a_2}\int_{b_1}^{b_2}\int_{c_1}^{c_2} f(x,y,z)\,dzdydx \\ &= \int_{c_1}^{c_2}\int_{a_1}^{a_2}\int_{b_1}^{b_2} f(x,y,z)\,dydxdz \\ &= \dots\end{aligned}$$

Multiplicatively separable integrals

- If the integrand is **multiplicatively separable**, i.e. it can be written as $f(x,y,z) = g(x)\,h(y)\,k(z)$, and the iterated integrals have constant limits of integration, then the triple integral can be calculated as a product of separate integrals

$$\int_{a_1}^{a_2}\int_{b_1}^{b_2}\int_{c_1}^{c_2} f(x,y,z)\,dzdydx$$
$$= \left(\int_{a_1}^{a_2} g(x)\,dx\right)\left(\int_{b_1}^{b_2} h(y)\,dy\right)\left(\int_{c_1}^{c_2} k(z)\,dz\right).$$

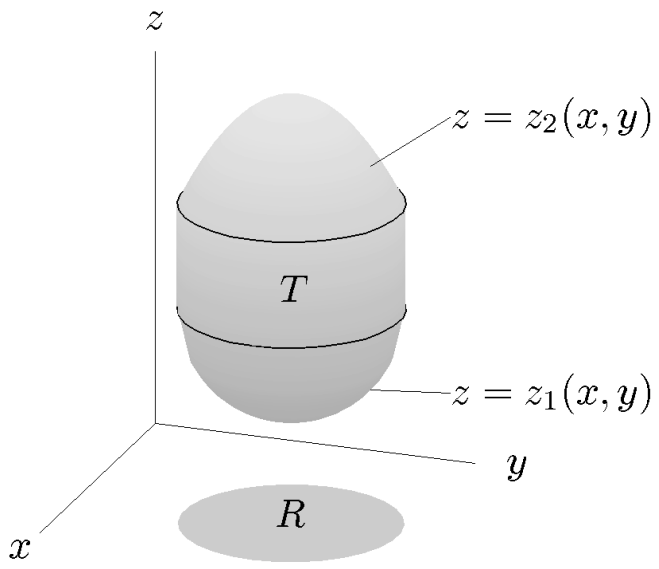
- Note that though the integrand is separable, but the integration region is not a cuboid, the triple integral cannot be calculated as separate integrals.

z-simple sets

- The set $T \subset \mathbb{R}^3$ is z-**simple**, if it is of the form

$$T = \left\{(x, y, z) \in \mathbb{R}^3 : (x, y) \in R, z_1(x, y) \leq z \leq z_2(x, y)\right\}.$$

where R is a set on the xy-plane that has area and z_1 and $z_2 : R \to \mathbb{R}$ are continuous functions.



Triple integral in a z-simple set

- Let $f : T \to \mathbb{R}$ be continuous. If T is z-simple, then

$$\iiint_T f(x,y,z)\,dxdydz = \iint_R \left(\int_{z_1(x,y)}^{z_2(x,y)} f(x,y,z)\,dz \right) dxdy.$$

  Similar results apply for y- and x-simple three dimensional sets.

- The limits of the inner integral can contain the variables of the middle and outer integrals, they are expressions of surfaces. The limits of the middle integral can contain the variable of outer integral, but not the inner. The outer integral has constant limits.

- Try to sketch a figure of the integration region, or at least its projections to the coordinate planes.

Properties of the triple integral

- A bounded set $T \subset \mathbb{R}^3$ has volume $v(T)$, if the constant function 1 is integrable in T. Then

$$v(T) = \iiint_T 1 \, dxdydz.$$

- If $a, b \in \mathbb{R}$ and f and g are integrable functions, then

$$\iiint_T (af + bg)(x, y, z) \, dxdydz$$
$$= a \iiint_T f(x, y, z) \, dxdydz + b \iiint_T g(x, y, z) \, dxdydz.$$

Mean value theorem of triple integrals

- The set $T \subset \mathbb{R}^3$ is **connected**, if any two points of T can be connected by a continuous curve contained entirely in T.
- Mean value theorem for triple integrals: Let $T \subset \mathbb{R}^3$ be closed, bounded and connected, and $v(T) > 0$. If $f : T \to \mathbb{R}$ is continuous, then there exists $(a, b, c) \in T$ such that

$$\iiint_T f(x, y, z)\, dxdydz = f(a, b, c)\, v(T).$$

- The **average value** of $f : T \to \mathbb{R}$, where $T \subset \mathbb{R}^3$, is

$$\bar{f} = \frac{1}{v(T)} \iiint_T f(x, y, z)\, dxdydz.$$

Representations of surface revolutions

- When replacing the variable x in certain equations of xz-plane curves with $\sqrt{x^2+y^2}$ creates a **surface of revolution** obtained by rotating the curve about the z-axis.
- line $z = x \Rightarrow$ cone $z = \sqrt{x^2+y^2}$
- parabola $z = x^2 \Rightarrow$ paraboloid $z = \left(\sqrt{x^2+y^2}\right)^2 = x^2+y^2$
- circle $x^2+z^2 = a^2 \Rightarrow$ sphere $x^2+y^2+z^2 = a^2$
- vertical line $x = a \Rightarrow x^2 = a^2 \Rightarrow$ cylinder $x^2+y^2 = a^2$

Cylinders

- If a three dimensional surface's equation does not contain the variable z, it is a cylinder in the direction of the z-axis. Its base is a shape on the xy-plane given by the same equation.
- circle of xy-plane $x^2 + y^2 = 9 \Rightarrow$ circular cylinder $x^2 + y^2 = 9$
- ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1 \Rightarrow$ elliptic cylinder $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$
- parabola $y = ax^2 \Rightarrow$ parabolic cylinder $y = ax^2$
- If a variable other than z is missing from the equation, we have a cylinder in the direction of the missing variable's axis.
- For example, the circular cylinder $y^2 + z^2 = 1$ is in the direction of the x-axis, with a circle of the yz-plane as its base.

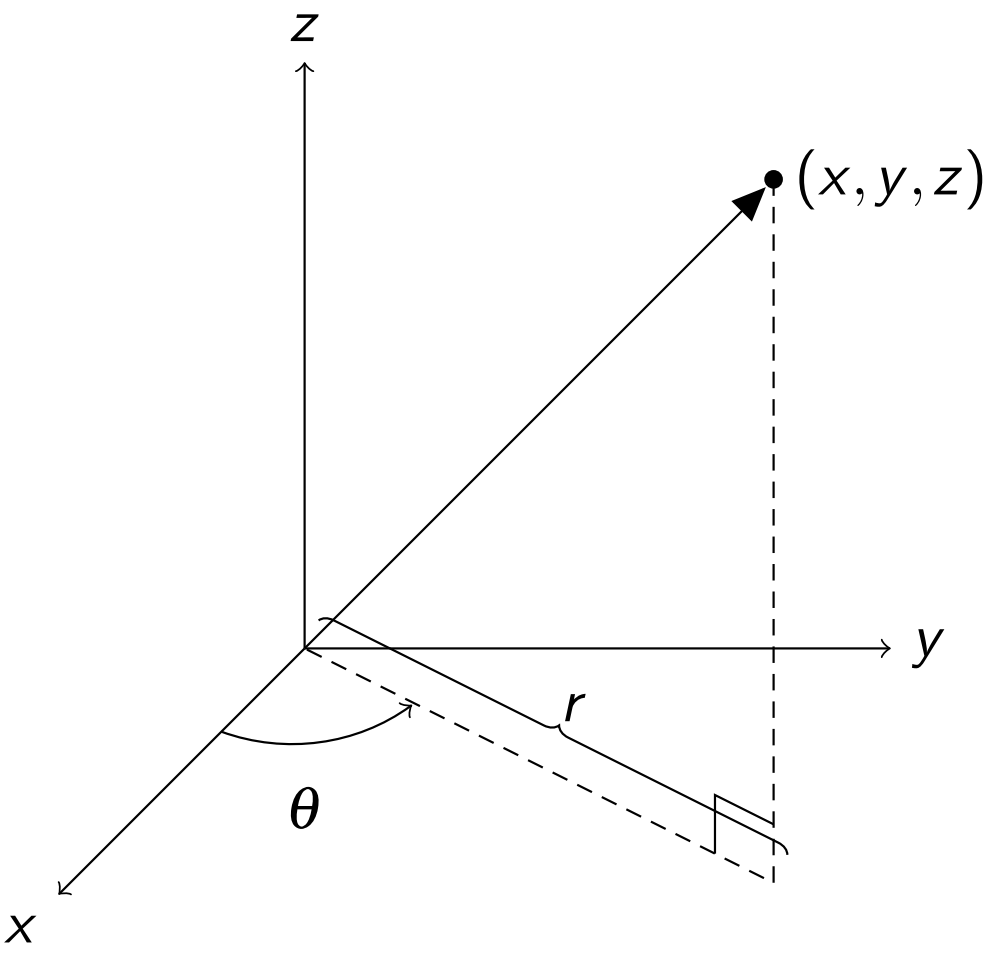
Cylindrical coordinates

- Consider $R \subset \mathbb{R}^2$, a projection of the z-simple set $T \subset \mathbb{R}^3$. If we express the points (x, y) of R in polar coordinates, we get the representation of (x, y, z) in **cylindrical coordinates** (r, θ, z).

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$$

- A cylindrical coordinate transformation is best used for sets whose boundary surface is a surface of revolution about the z-axis, or a part of a set cut in a certain way (for example cut along a plane through the origin and in the direction of the z-axis).

Cylindrical coordinates graphically



$$r = \sqrt{x^2 + y^2} \geq 0,$$
$$\theta \in [0, 2\pi]$$
$$\text{or } \theta \in [-\pi, \pi],$$
$$z \in \mathbb{R}.$$

Cylindrical coordinate transformation in an integral

- Let $T \subset \mathbb{R}^3$ be a z-simple set, $f : T \to \mathbb{R}$ continuous and $(r, \theta, z) \in U$ the cylindrical coordinates for the point $(x, y, z) \in T$. Then

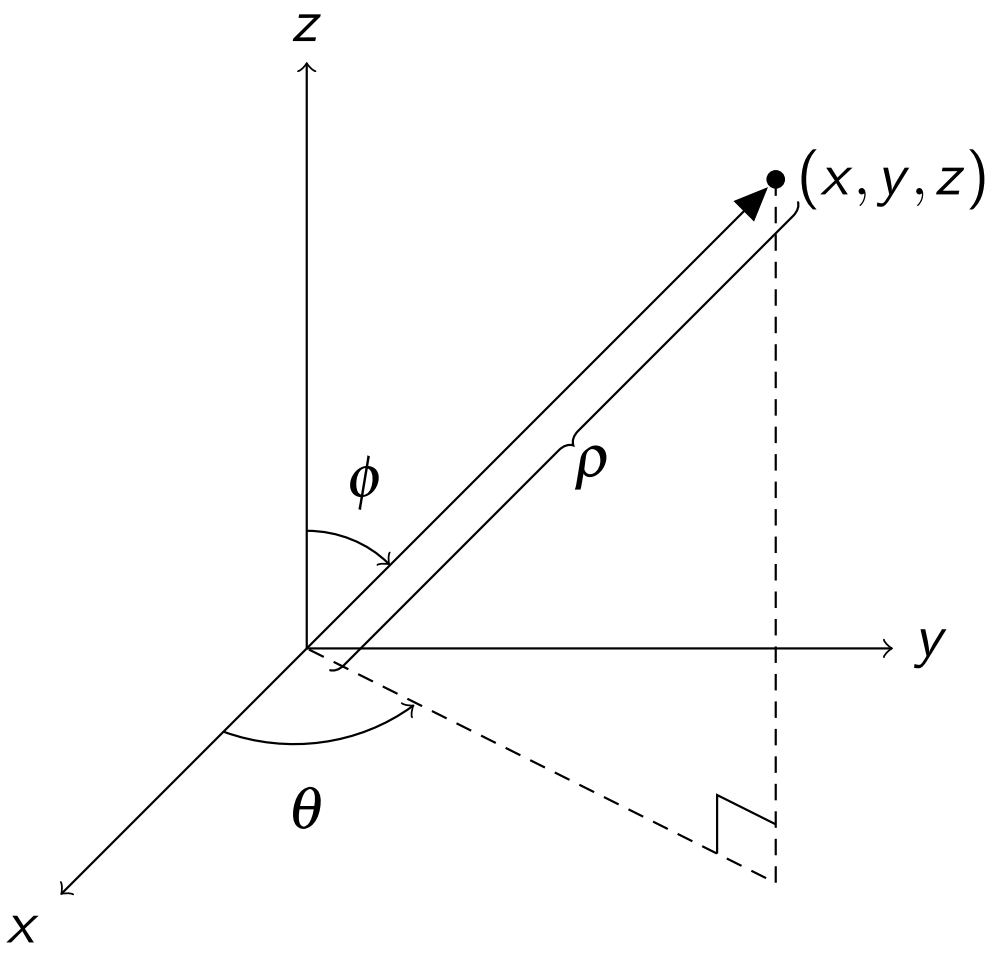
$$\iiint_T f\, dxdydz = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} \int_{z_1(r,\theta)}^{z_2(r,\theta)} f\left(r\cos\theta, r\sin\theta, z\right) r\, dzdrd\theta.$$

- Note three things about the transformation:
  - substitute $x = r\cos\theta$, $y = r\sin\theta$, $z = z$ in the function f,
  - multiply the integrand by the Jacobian determinant r,
  - find the limits of U in cylindrical coordinates by sketching the set.

Spherical coordinates

- Denote the length of (x, y, z) by ρ (rho), the angle between the positive z-axis and the vector (x, y, z) by ϕ (phi) and the angle between the positive x-axis and the projection vector $(x, y, 0)$ by θ (theta).
- We get the coordinate representation (ρ, ϕ, θ), where
  - θ is the same angle as for cylindrical coordinates,
  - ρ is not equal to r of the cylindrical coordinates: ρ is the distance from the origin, r is the distance from the z-axis,
  - ϕ is an angle from the positive z-axis (where it is zero) to the direction determined by θ, and ending no further than the negative z-axis (where it gets the value π). Note that ϕ will <u>never</u> rotate upwards after reaching the negative z-axis.

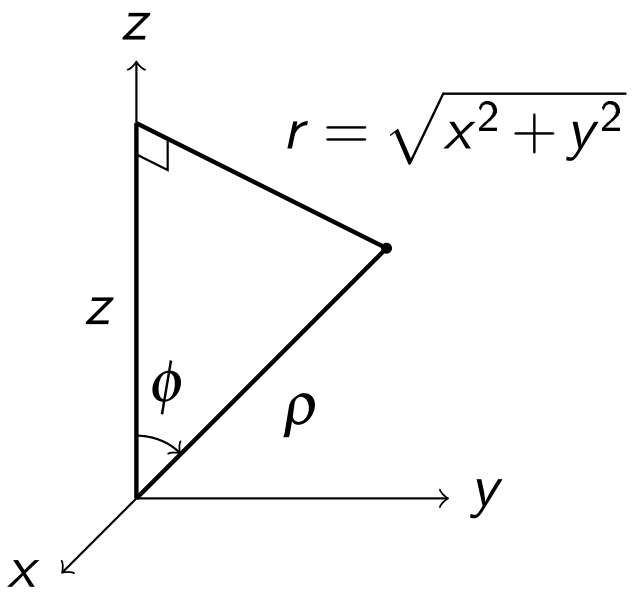
Spherical coordinates graphically



$$\rho = \sqrt{x^2 + y^2 + z^2} \geq 0,$$
$$\phi \in [0, \pi],$$
$$\theta \in [0, 2\pi]$$
$$\text{or } \theta \in [-\pi, \pi].$$

Connection to cartesian and cylindrical coordinates

- The triangle in the figure gives a representation to r and z.



$$r = \rho \sin \phi$$

and

$$z = \rho \cos \phi$$

- By the cylindrical coordinates we additionally know that

$$x = r \cos \theta = \rho \sin \phi \cos \theta \quad \text{and} \quad y = r \sin \theta = \rho \sin \phi \sin \theta.$$

Spherical coordinate transformation

- The triplet (ρ, ϕ, θ) is called the **spherical coordinates**.

$$\begin{cases} x = \rho \sin\phi \cos\theta \\ y = \rho \sin\phi \sin\theta \\ z = \rho \cos\phi \end{cases}$$

- θ - symbol with horizontal line - angle rotating horizontally, ϕ - symbol with vertical line - angle increasing downwards.
- Warning: In physics they use an ISO standardized version of the spherical coordinates with the symbols of the angles interchanged.

Spherical coordinate transformation in an integral

- Let $T \subset \mathbb{R}^3$ be a set (that is "nice enough"), $f : T \to \mathbb{R}$ continuous and $(\rho, \phi, \theta) \in U$ the cylindrical coordinates for the point $(x, y, z) \in T$. Then

$$\iiint_T f(x, y, z)\, dxdydz$$
$$= \iiint_U f(\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi) \cdot \rho^2 \sin\phi \, d\rho\, d\phi\, d\theta.$$

- The Jacobian determinant is now $\rho^2 \sin\phi$.
- The spherical coordinate transformation is usually used in integrals for sets like a ball or an object cut from a ball.

Mass and center of mass

- The **mass** of an object $T \subset \mathbb{R}^3$, with density $\rho(x, y, z)$, is

$$m = \iiint_T \rho(x, y, z)\, dxdydz.$$

- The coordinates of the **center of mass** $(\bar{x}, \bar{y}, \bar{z})$ of T can be calculated as

$$\bar{x} = \frac{1}{m} \iiint_T x\rho\, dV, \quad \bar{y} = \frac{1}{m} \iiint_T y\rho\, dV, \quad \bar{z} = \frac{1}{m} \iiint_T z\rho\, dV.$$

- The **moment of inertia** of T with respect to the z-axis is

$$I_z = \iiint_T \left(x^2 + y^2\right) \rho(x, y, z)\, dV.$$

Change of variables transformation, two variables

- A **change of variables transformation** is the function

$$F : \mathbb{R}^2_{uv} \to \mathbb{R}^2_{xy}, \qquad F(u,v) = (x(u,v), y(u,v)),$$

transforming a point given in uv-coordinates to xy-coordinates.

- The Jacobian determinant connected to this change of variables is

$$J_F(u,v) = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u}(u,v) & \frac{\partial x}{\partial v}(u,v) \\ \frac{\partial y}{\partial u}(u,v) & \frac{\partial y}{\partial v}(u,v) \end{vmatrix}.$$

Change of variables in a double integral

- Let A be a "nice enough" set, $f : A \to \mathbb{R}$ a continuous function, a continuously differentiable change of variables transformation F with $F(A) = S$ and $J_F(u,v) \neq 0$ in the interior of S.
- The integral we get after changing the variables is

$$\iint_A f(x,y)\,dxdy = \iint_S f(F(u,v)) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| dudv.$$

- Note that, for example, for the spherical coordinates it holds that $\rho \geq 0$ and $0 \leq \phi \leq \pi$, so $J_F(\rho,\phi,\theta) = \rho^2 \sin\phi$ is always non-negative. In a general change of variables, remember to multiply by the absolute value of the Jacobian determinant $|J_F(u,v)|$.

Alternative way to find the Jacobian determinant

- For a change of variables of the form $x = x(u, v)$ and $y = y(u, v)$, the Jacobian determinant is calculated "as is".
- For a change of variables $u = u(x, y)$ and $v = v(x, y)$, you either first find representations for x and y with respect to u and v, or you can use the information

$$\frac{\partial(x, y)}{\partial(u, v)} = \left(\frac{\partial(u, v)}{\partial(x, y)}\right)^{-1}.$$

- However, remember that the changed integrand can only depend on the variables u and v (if it doesn't after the transformation, find and substitute the representations of x and y with respect to u and v).

Change of variables in a triple integral

- The Jacobian determinant connected to this change of variables is

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial x}{\partial u}(u,v,w) & \frac{\partial x}{\partial v}(u,v,w) & \frac{\partial x}{\partial w}(u,v,w) \\ \frac{\partial y}{\partial u}(u,v,w) & \frac{\partial y}{\partial v}(u,v,w) & \frac{\partial y}{\partial w}(u,v,w) \\ \frac{\partial z}{\partial u}(u,v,w) & \frac{\partial z}{\partial v}(u,v,w) & \frac{\partial z}{\partial w}(u,v,w) \end{vmatrix}.$$

- The integral we get after changing the variables is

$$\iiint_A f(x,y,z)\,dxdydz = \iiint_S f(F(u,v,w)) \left| \frac{\partial(x,y,z)}{\partial(u,v,w)} \right| dudvdw.$$

Improper integrals

- In previous integrations we assumed that the region A was bounded and that the function f was bounded (i.e. stays finite even when approaching the boundary of the region).
- Next, define integrals for an unbounded set of integration and for an unbounded integrand in double integration. Triple integrals would be defined similarly.
- The definitions contain a limit. If a finite limit exists, the integral **converges**. If the limit does not exists because of oscillation or the limit is infinite, the integral **diverges**.

Unbounded set of integration

- Let A be a "nice enough" unbounded set and $f : A \to \mathbb{R}$ a bounded function such that $f(x,y) \geq 0$ for all $(x,y) \in A$.
- **Cover** A with an increasing sequence of bounded subsets A_k,

$$A_1 \subset A_2 \subset A_3 \subset \cdots \quad \text{and} \quad A = \bigcup_{k=1}^{\infty} A_k = A_1 \cup A_2 \cup A_3 \cup \cdots$$

- If f is integrable in each A_k, define

$$\iint_A f(x,y)\, dxdy = \lim_{k\to\infty} \iint_{A_k} f(x,y)\, dxdy.$$

- The idea: Evaluate the integral in a bounded set and extend the result to an unbounded set by taking limit.

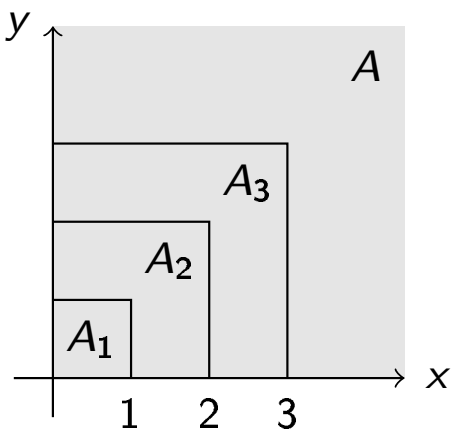
An example of an unbounded set of integration

- Let $A = \{(x,y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\}$. Evaluate the integral

$$I = \iint_A \frac{1}{(1+x^2)(1+y^2)} dxdy.$$

- The integrand is bounded, $0 < f(x,y) \leq 1$, but A not. Choose

$$A_k = \{(x,y) \in \mathbb{R}^2 : 0 \leq x \leq k, 0 \leq y \leq k\}.$$



An example of an unbounded set of integration

- Evaluate the integral using the chosen sequence (A_k).

$$
\begin{aligned}
I &= \lim_{k\to\infty} \iint_{A_k} \frac{1}{(1+x^2)(1+y^2)}\,dxdy \\
&= \lim_{k\to\infty} \int_0^k \int_0^k \frac{1}{(1+x^2)(1+y^2)}\,dxdy \\
&= \lim_{k\to\infty} \left(\int_0^k \frac{1}{1+x^2}\,dx\right)\left(\int_0^k \frac{1}{1+y^2}\,dy\right) \\
&= \lim_{k\to\infty} \Big[\arctan x\Big]_{x=0}^{x=k} \cdot \Big[\arctan y\Big]_{y=0}^{y=k} \\
&= \left(\lim_{k\to\infty}(\arctan k - \underbrace{\arctan 0}_{=0})\right)^2 = \left(\frac{\pi}{2}\right)^2 = \frac{\pi^2}{4}.
\end{aligned}
$$

Unbounded integrand

- Let A be a "nice enough" bounded set and $f : A \to \mathbb{R}$ an unbounded function such that $f(x,y) \geq 0$ for all $(x,y) \in A$.
- Cover A with an increasing sequence of bounded subsets A_k,

$$A_1 \subset A_2 \subset A_3 \subset \cdots \quad \text{and} \quad A = \bigcup_{k=1}^{\infty} A_k.$$

- If f is integrable in each A_k, define

$$\iint_A f(x,y)\,dxdy = \lim_{k\to\infty} \iint_{A_k} f(x,y)\,dxdy.$$

- The idea: In A_k we isolate the set where f is unbounded from A. Extend the value of the integral in A_k to A by taking limit.

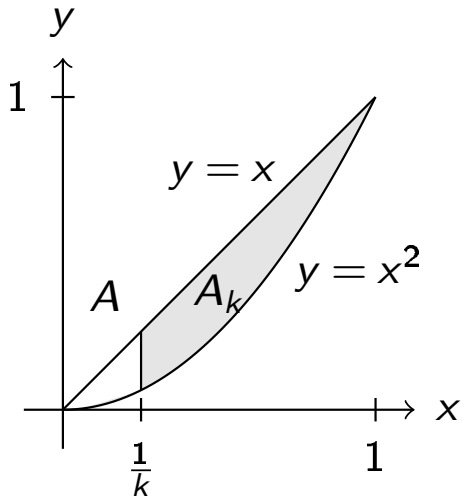
An example of integrating an unbounded function

- Let A be bounded by the curves $y = x$ and $y = x^2$. Evaluate

$$I = \iint_A \frac{1}{xy} dxdy.$$

- The set A is bounded, but $\lim\limits_{(x,y)\to(0,0)} f(x,y) = \infty$. Choose

$$A_k = \left\{ (x,y) \in \mathbb{R}^2 : x^2 \le y \le x, \frac{1}{k} \le x \le 1 \right\}.$$



An example of integrating an unbounded function

- Evaluate the integral using the chosen sequence (A_k).

$$I = \lim_{k\to\infty} \int_{\frac{1}{k}}^{1} \frac{1}{x} \int_{x^2}^{x} \frac{1}{y} dy dx = \lim_{k\to\infty} \int_{\frac{1}{k}}^{1} \frac{1}{x} \Big[\ln y\Big]_{y=x^2}^{y=x} dx$$

$$= \lim_{k\to\infty} \int_{\frac{1}{k}}^{1} \frac{1}{x} \underbrace{\left(\ln x - \ln x^2\right)}_{=\ln x - 2\ln x = -\ln x} dx = -\lim_{k\to\infty} \int_{\frac{1}{k}}^{1} \underbrace{\frac{1}{x}}_{=g'} \cdot (\underbrace{\ln x}_{=g})^1 \, dx$$

$$= -\lim_{k\to\infty} \left[\frac{1}{2}(\ln x)^2\right]_{x=\frac{1}{k}}^{x=1} = -\frac{1}{2} \lim_{k\to\infty} \left((\underbrace{\ln 1}_{=0})^2 - \left(\ln \tfrac{1}{k}\right)^2\right)$$

$$= \lim_{k\to\infty} \frac{\left(\ln k^{-1}\right)^2}{2} = \lim_{k\to\infty} \frac{(-\ln k)^2}{2} = \lim_{k\to\infty} \frac{(\ln k)^2}{2} = \infty.$$